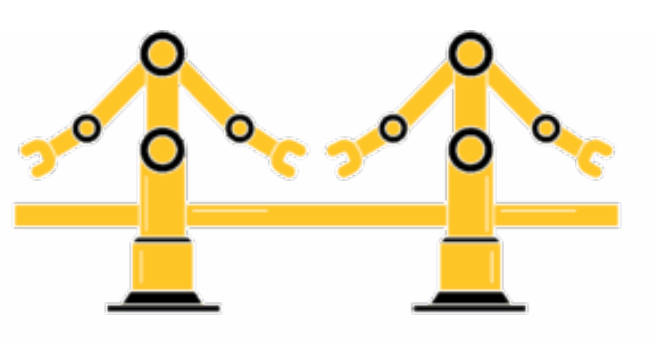




Multi-digit Force Synergies for Haptic Interface Design

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Introduction

In-hand manipulation is essential for dexterous teleoperation and immersive interaction in virtual environments¹. While existing haptic interfaces mainly focus on static grasping, their extension to in-hand manipulation is restricted by current limited investigation of the underlying haptic dimensions². Although manipulation kinematics have been extensively studied³, fingertip force patterns remain relatively unexplored. To address this gap, we developed a customized setup to measure finger forces during in-hand manipulation and investigate their underlying force patterns using Principal component analysis (PCA)

Method

A. Participants

Two laboratory members participated in the preliminary user study. For the formal study, we plan to recruit **10-15** healthy unimpaired, right handed subjects through flyers.

B. Equipment

We developed a free-hand manipulation setup in which users manipulate customized cylindrical objects of varying diameters. Each object integrates three load cells to measure the forces exerted by the thumb, index finger, and middle finger, along with an electromagnetic tracker for measuring the object pose.

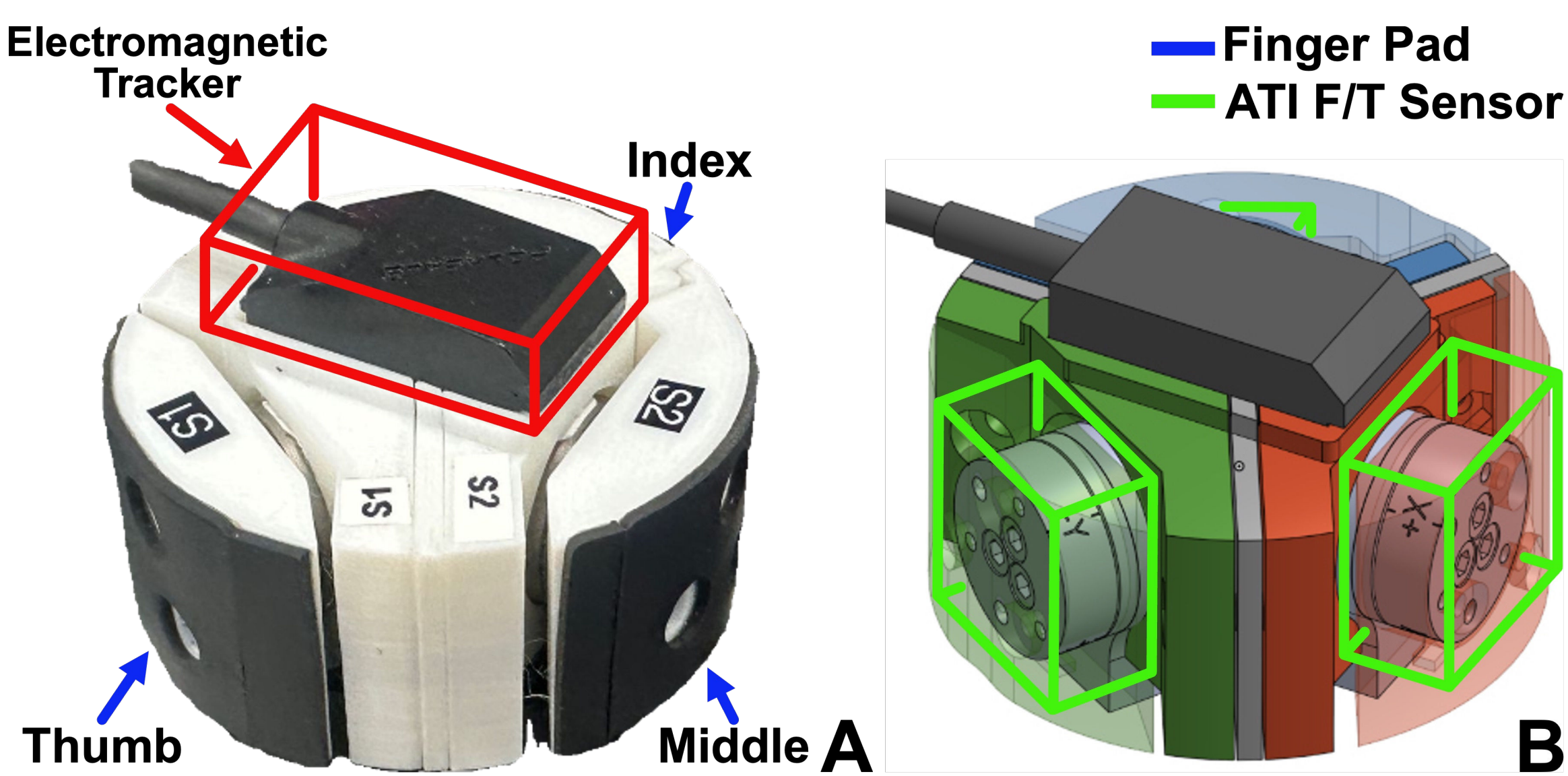


Fig. 1. Customized cylindrical object and its CAD design. (A) Physical prototype with three finger-contact regions for the thumb, index finger, and middle finger. (B) CAD design illustrating the configuration and placement of the three embedded load cells.

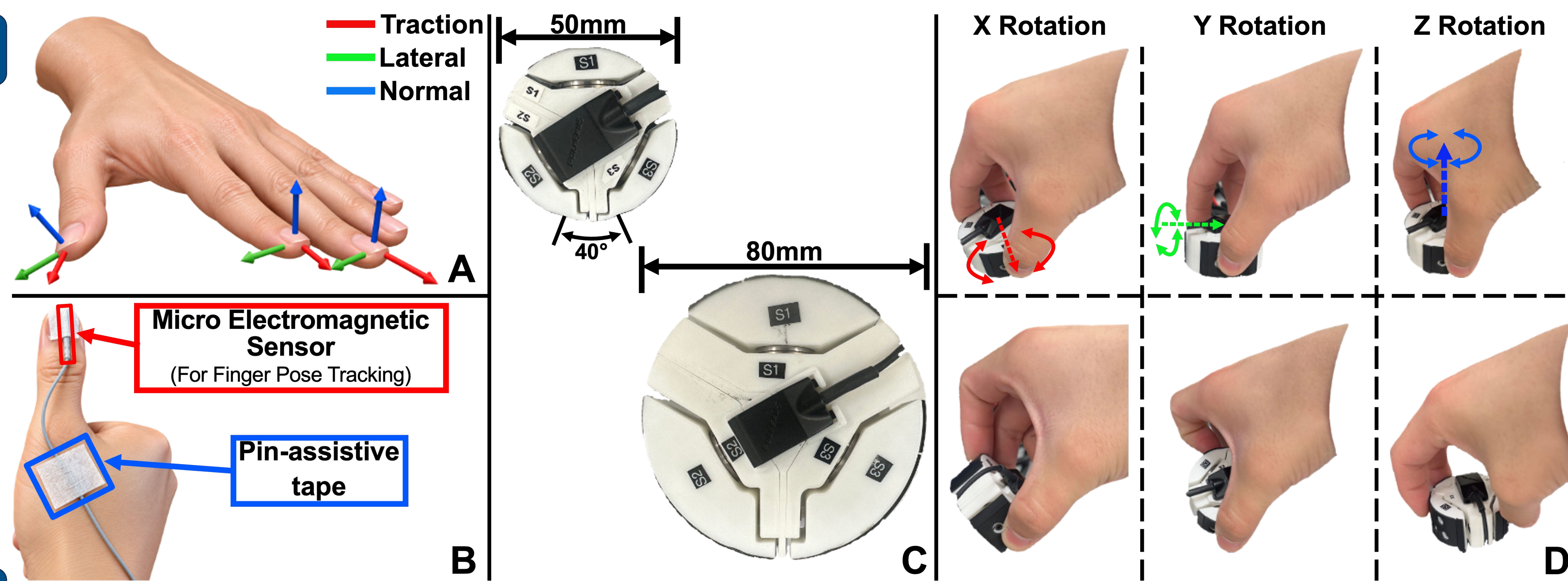


Fig. 2. Free-hand manipulation setup for finger-force and pose measurement. (A) Finger-based force coordinate system, with traction, lateral, and normal force components shown in red, green, and blue, respectively. (B) Wearable electromagnetic sensor attached to the fingertip for finger-pose tracking. (C) Customized cylindrical objects with different diameters (50 and 80 mm) and a 40° opening between adjacent finger-contact regions. (D) Representative free-hand manipulation tasks involving rotations about the X, Y, and Z axes.

C. Procedure

The study includes **four free-manipulation tasks**: rotations about the X, Y, and Z axes, and **finger gaing**. Due to constraints imposed by the placement of the load cells, the finger-gaiting task is limited to alternating back-and-forth movements of the index and middle fingers, while the thumb remains in contact with its designated sensing region. Each tasks will be perform five trials using two customized cylindrical objects with diameter of 50 and 80 mm. Breaks will be given between trials, tasks, and objects, as well as upon participants request, to minimize fatigue. The complete study is expected to take approximately **2 hours** per participant.

D. Data Processing and Analysis

During the experiments, **nine** force signals will be recorded, corresponding to the traction, lateral, and normal force components for each finger. Principal component analysis (PCA) will be performed on the **nine-dimensional force data** for each trial to identify the dominant patterns of force coordination among the three fingers. The variability of the PCA results across repeated trials of the same task will then be quantified using the standard deviation to assess trial-to-trial consistency and repeatability.

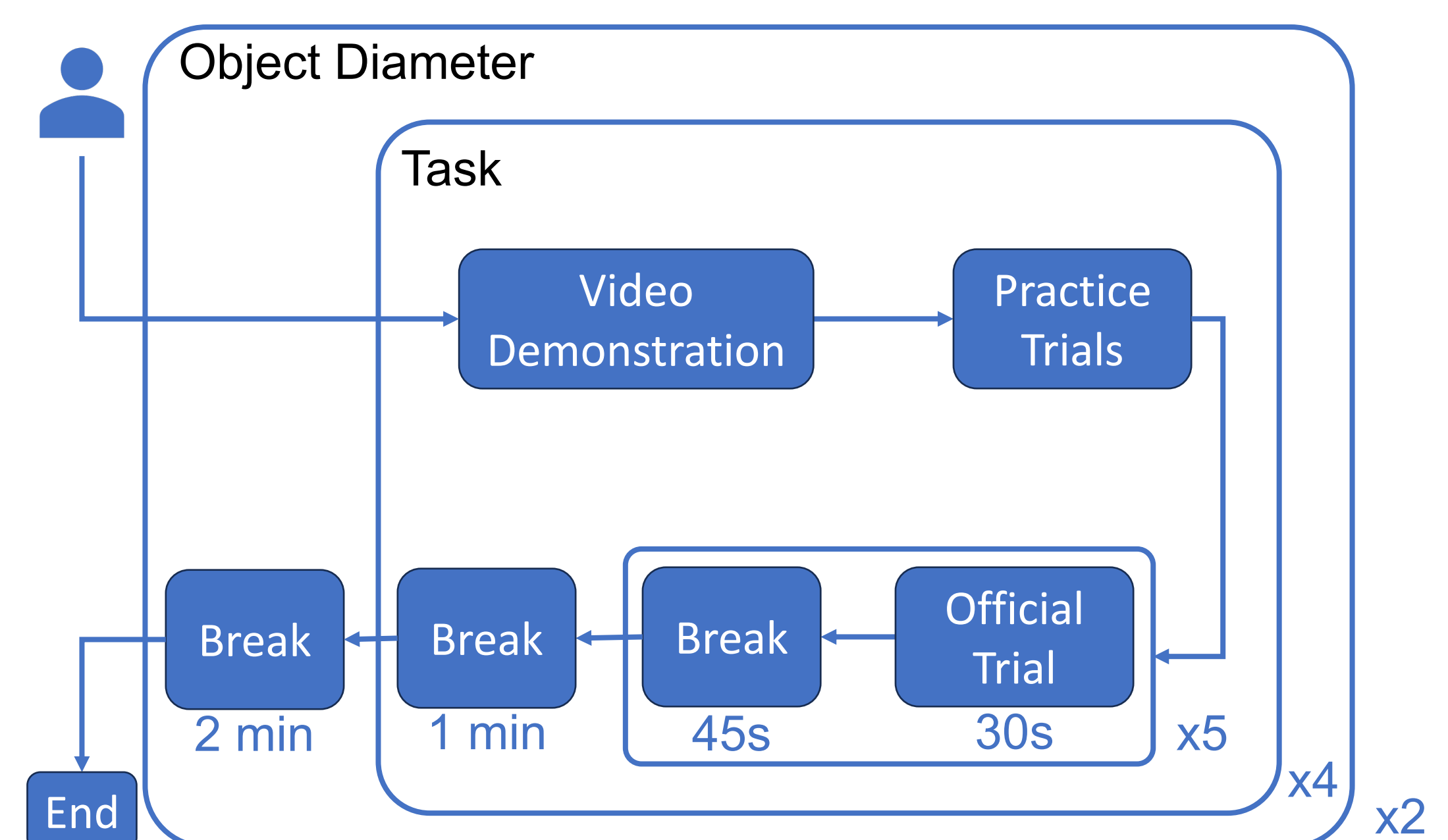


Fig. 3. Experimental protocol for the in-hand manipulation study.

Preliminary Data

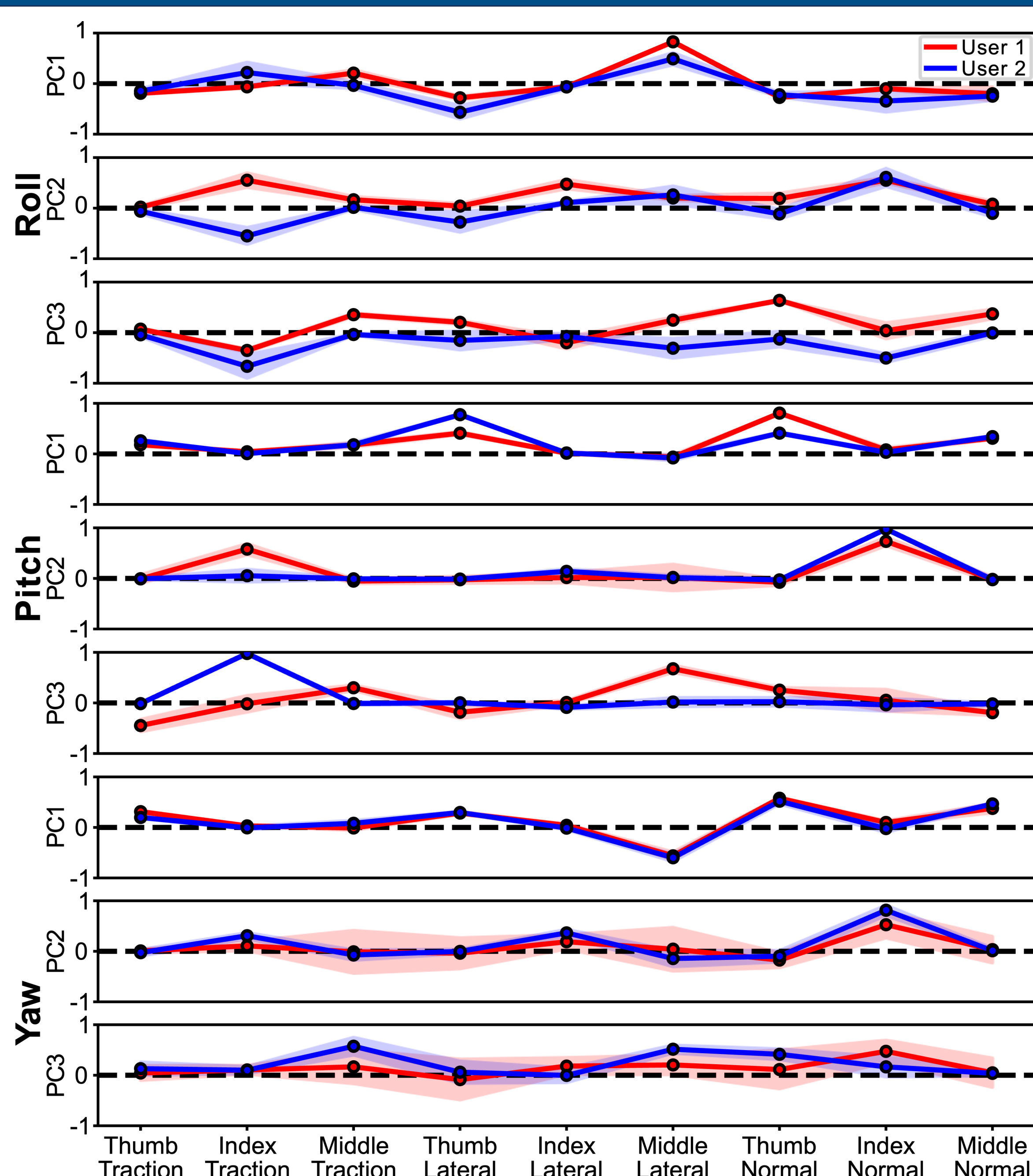


Fig. 4. Principal component (PC) patterns of finger forces for two participants during roll, pitch, and yaw tasks. PC1–PC3 are shown for each task. The nine dimensions correspond to the traction, lateral, and normal force components of the thumb, index, and middle fingers. Solid curves show the mean PC loadings across repeated trials for User 1 (red) and User 2 (blue), with shaded regions indicating the standard deviation.

Discussion

Preliminary Findings

Preliminary measurements support the following key findings:

- Demonstrate the functionality and feasibility of the current experimental setup.
- Suggest distinct multi-digit force correlations across different manipulation motions.

Limitations and Improvements

The preliminary study also highlighted several areas for improvement:

- Clearer and more straightforward instructions should be incorporated into the video demonstration.
- Quantitative movement-error feedback should be provided during practice trials, as verbal instructions alone were insufficient for participants to consistently perform perpendicular rotational movements.

Future Work:

- Standardize and refine task instructions to improve consistency across participants.
- Integrate real-time movement-error feedback to guide participants during practice trials.

References

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